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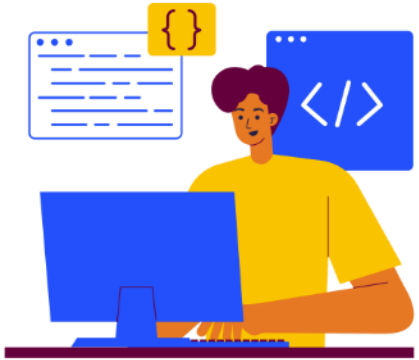
SINGAPORE INSTITUTE OF TECHNOLOGY

Full Stack Secured Development

Node.js Web API



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Learning Outcomes

By the end of this lesson, you will be able to:

1. Explain the role of Node.js in Full Stack Development.
2. Define what a RESTful API is.
3. Discuss how to use Express.js for building web APIs.
4. Outline some useful packages.

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Node.js

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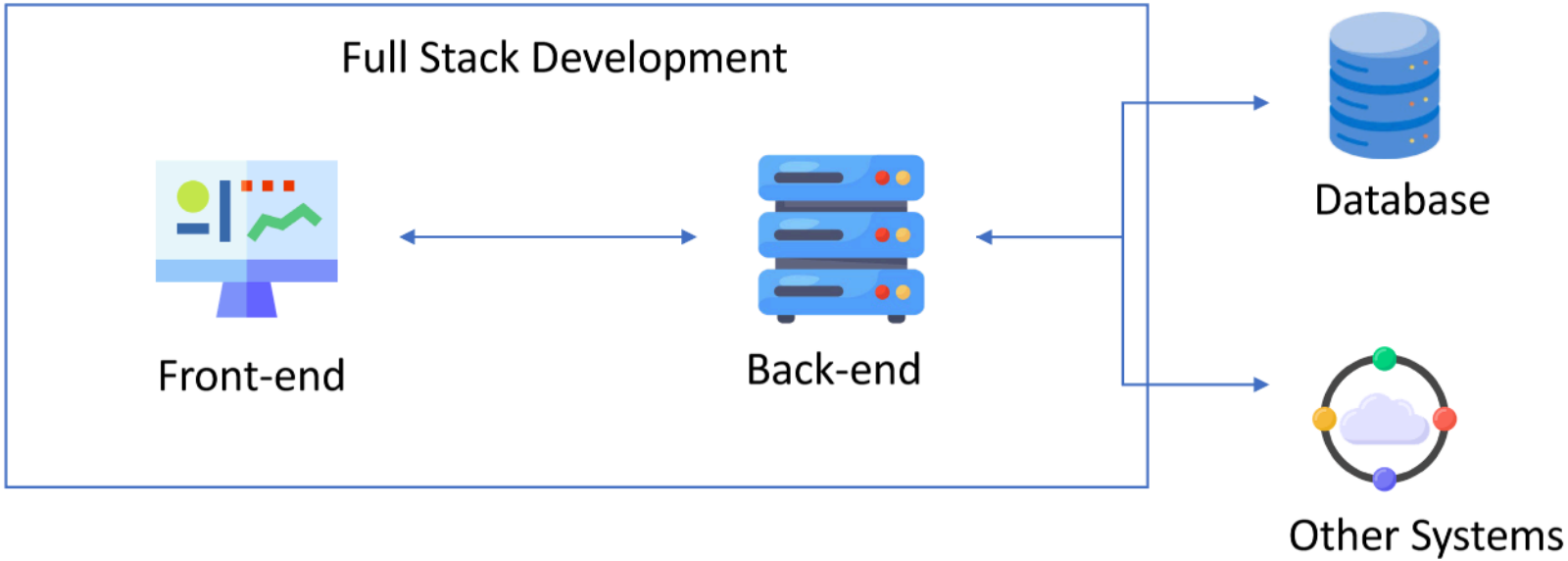
Full Stack Development

Full stack development refers to the end-to-end web application development, including the **front-end** and **back-end**.



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Node.js for Back-end

Node.js is an open-source, server-side JavaScript runtime environment.

- **Asynchronous**: uses an event-driven, non-blocking I/O model
- **Scalability**: can handle a large number of requests and processing tasks
- **Speed**: popular choice for building high-performance web applications
- **Cross-platform**: can run on different operating systems

```
node -v  # check version
```

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Hello Node.js

Simple code example to print "Hello, Node.js!" to the console

app.js

```
console.log("Hello, Node.js!");
```

- Create a JavaScript file

Open terminal or command prompt

```
> cd /path/to/your/directory
```

- Navigate to the file directory

```
> node app.js
```

- Run Node.js script

Output:

```
Hello, Node.js!
```

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npm

npm (Node Package Manager) is a tool for managing JavaScript packages in Node.js projects. It is bundled with Node.js by default.

- npm init -y # -y: use default options
- npm install # shorthand: “npm i”
- npm install <package>
- npm install <package>@<version>
- npm install -g <package> # -g: install globally
- npm install --save-dev <package> # --save-dev: save as development dependency
- npm uninstall <package>
- npm update
- npm run <script>

npm -v # check version

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package.json

```
{
  "name": "my-node-app",
  "version": "1.0.0",
  "description": "A simple Node.js application",
  "main": "index.js",
  "dependencies": {
    "express": "^4.18.2"
  },
  "devDependencies": {
    "nodemon": "^2.0.13"
  },
  "scripts": {
    "start": "node index.js",
    "dev": "nodemon index.js"
  },
  "author": "Your Name",
  "license": "MIT"
}
```

- The package.json file is a JSON file that serves as the manifest for Node.js projects.
- It contains metadata about the project, including its name, version, dependencies, scripts, and other important information.

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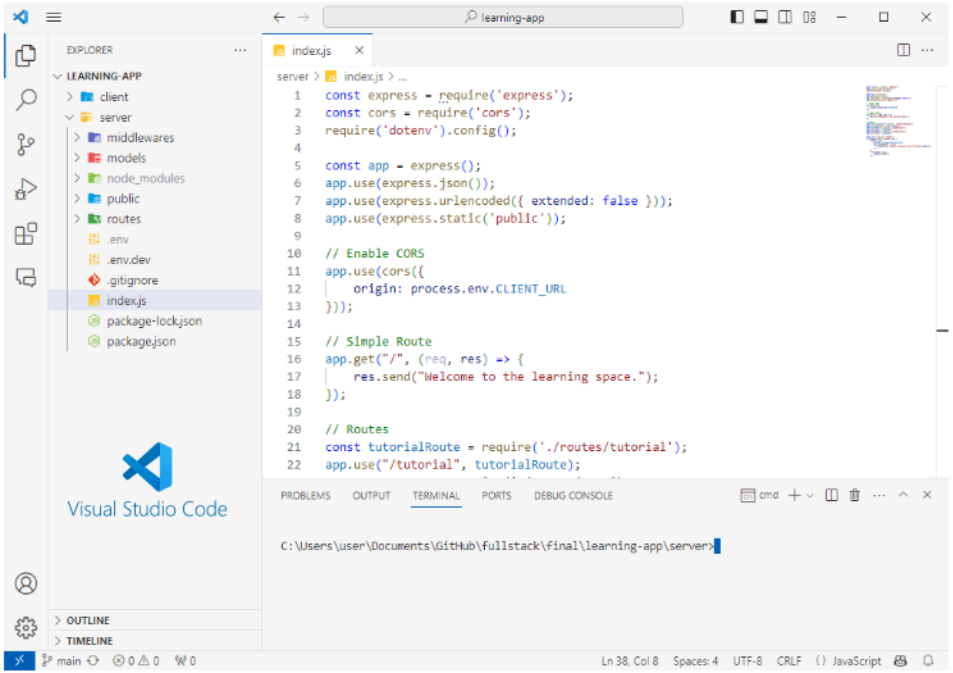
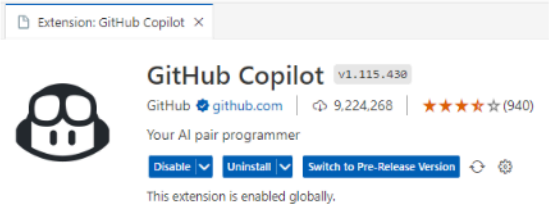
Code Editor

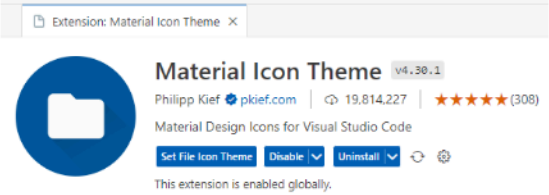
- Visual Studio Code (VS Code) is a popular code editor. It includes IntelliSense for Node.js by default.



Useful extensions:



- Help you to code faster, free for students



- Nice colourful icons for folders and files

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RESTful API

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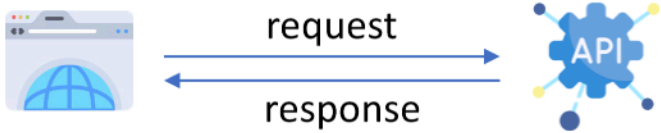
RESTful API

A RESTful API, or Representational State Transfer API, is a set of principles for designing and structuring web services.

- **Resource-Based:** identify the resources that the API exposes (e.g. user, product)
- **CRUD Operations:** CRUD stands for Create, Read, Update, and Delete
- **Uniform Interface:** use HTTP methods, URLs and JSON data
- **Statelessness:** not rely on any previous request or server session



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HTTP Methods

HTTP (Hypertext Transfer Protocol) defines a set of request methods that are used by the client to send requests to the server.

- GET: **read** data from resource(s)
- POST: **create** a new resource
- PUT: **update** an existing resource
- DELETE: **delete** a resource

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Routes and Endpoints

- **Route**: A route is a path that defines a specific resource or a group of related resources within the API. (e.g. “/product”)

- **Endpoint**: An endpoint is a specific URL pattern within a route, and it's associated with an HTTP method.
  - GET /product
  - GET /product/{id}
  - POST /product
  - PUT /product/{id}
  - DELETE /product/{id}

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HTTP Status Codes

HTTP uses response status codes to indicate the result of a client's request to the server.

- **200 OK**: The request was successful, and the server has returned the requested data
- **400 Bad Request**: Invalid syntax or missing information in the request
- **401 Unauthorized**: The client is not authorized to access the requested resource
- **403 Forbidden**: The client does not have permission to access the requested resource
- **404 Not Found**: The requested resource could not be found on the server
- **500 Internal Server Error**: The server encountered an unexpected error

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JSON for Data

JSON (JavaScript Object Notation) is a lightweight data interchange format that is commonly used in API request and response bodies.

JSON for one user

```
{
    "name": "John Doe",
    "email": "john.doe@example.com",
    "age": 30
}
```

JSON for list of users

```
{
    "users": [
        {
            "name": "John Doe",
            "email": "john.doe@example.com",
            "age": 30
        },
        {
            "name": "Jane Smith",
            "email": "jane.smith@example.com",
            "age": 25
        }
    ]
}
```

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Popular Frameworks

- Express.js (Node.js)
- ASP.NET Core (C#)
- Flask (Python)
- Spring Boot (Java)

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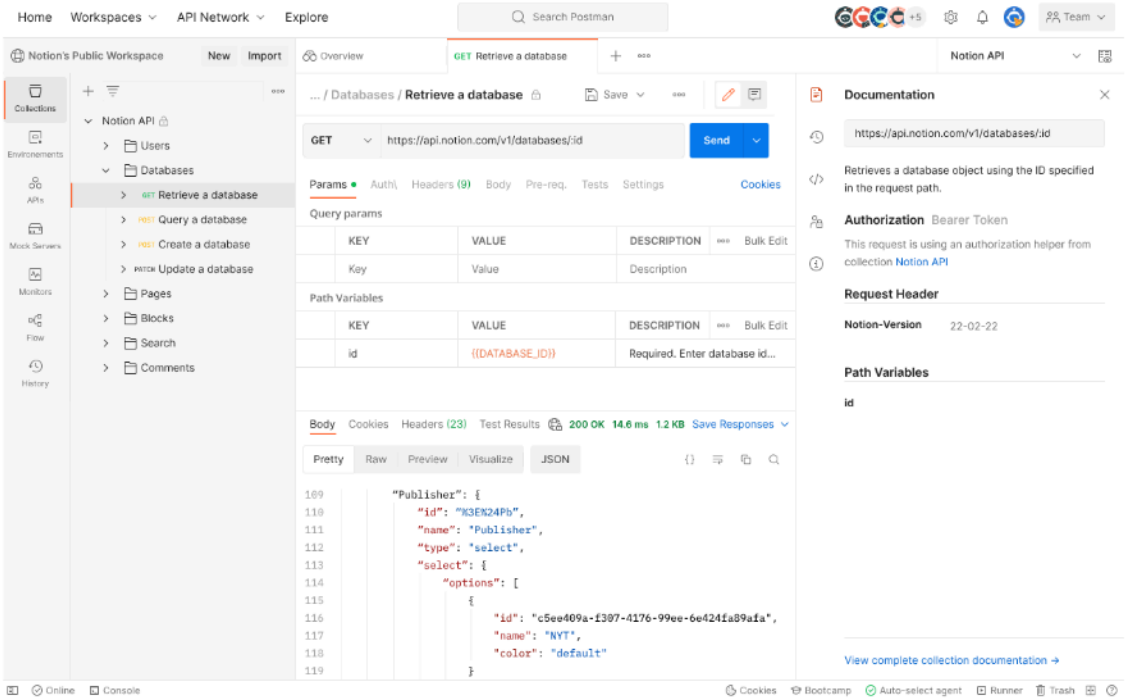
Testing API

Postman is a popular tool for testing web APIs.

- Saving requests
- Team collaboration



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Express.js

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Express.js

Express.js is a popular Node.js framework that provides tools and features for building RESTful APIs.

npm i express # install express

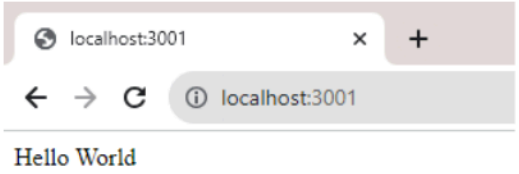
index.js

```
const express = require('express');
const app = express();

app.get('/', function (req, res) {
  res.send('Hello World');
})

let port = 3001;
app.listen(port, () => {
    console.log('Sever running');
});
```

- Import express and create an application
- Define a route handler for the root URL
- Start the server



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Project Structure

- project-root
  - > middlewares (custom middleware functions)
  - > models (database models)
  - > node_modules (libraries and dependencies generated by npm)
  - > routes (API route definition)
  - .env (environment variables)
  - index.js (entry point of the application)
  - package-lock.json (lock file generated by npm)
  - package.json (project configuration including dependencies)

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Create Route

routes/product.js

```js
const express = require('express');
const router = express.Router();

const products = [
    { id: 1, name: 'Product 1', price: 10.99 },
    { id: 2, name: 'Product 2', price: 19.99 }
  ];

// Define a GET endpoint to list all products
router.get('/', (req, res) => {
    res.json(products);
});

module.exports = router;
```

- Create the route file
- Import express and create an express router
- Define a list of products
- Define a GET endpoint to list all products
- Export the router

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Use Route

index.js

```
// Import the route
const productRoute = require('./routes/product');

// Mount the route on a specific URL path
app.use('/product', productRoute);
```

- './routes/product' specifies the relative path to the product.js file
- All the endpoints defined in the product route will be under the URL path '/product'

route path '/product' + get('/', ...) → endpoint 'GET /product': list all products

This approach allows you to organize your application's routes into modular files and specify where those routes should be accessible.

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POST

endpoint ‘POST /product’: create a new product

routes/product.js

```
// Define a POST endpoint to create a new product
router.post("/", (req, res) => {
    let data = req.body;
    products.push(data);
    res.json(data);
});
```

- Extract data from the request body
- Add data into the array
- Return the data

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GET by ID

endpoint ‘GET /product/{id}’: get a product by ID

routes/product.js

```
// Define a GET endpoint to get a product by ID
router.get('/:id', (req, res) => {
    const id = req.params.id;
    const product = products.find((p) => p.id === id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    res.json(product);
});
```

- Extract parameter id from URL
- Check the error Not Found
- Return the found data

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PUT by ID

endpoint ‘PUT /product/{id}’: update a product by ID

routes/product.js

```
// Define a PUT endpoint to update a product by ID
router.put('/:id', (req, res) => {
    const id = req.params.id;
    const product = products.find((p) => p.id === id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    // TODO: Update the product
    res.json({ message: 'Product updated' });
});
```

- Extract parameter id from URL
- Check the error Not Found
- Update data

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DELETE by ID

endpoint ‘DELETE /product/{id}’: delete a product by ID

routes/product.js

```
// Define a DELETE endpoint to delete a product by ID
router.delete('/:id', (req, res) => {
    const id = req.params.id;
    const product = products.find((p) => p.id === id);
    if (!product) {
        res.sendStatus(404);
        return;
    }
    // TODO: Delete the product
    res.json({ message: 'Product deleted' });
});
```

- Extract parameter id from URL
- Check the error Not Found
- Delete data

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Useful Packages

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nodemon

nodemon is a tool that helps develop Node.js based applications by automatically restarting the node application when file changes in the directory are detected.

- It is particularly useful during the development process, as it eliminates the need to manually stop and restart your application every time you make a code change.
- nodemon is often used in combination with npm scripts. You can define a script in your package.json file to run your application with Nodemon.

npm i --save-dev nodemon

package.json

```
"scripts": {
  "start": "nodemon index.js"
}
```

- When you run "**npm start**", npm will execute the "**nodemon index.js**" command

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dotenv

dotenv package is used to load environment variables from a .env file into the process.env object.

npm i dotenv

.env

```
APP_PORT = 3001
```

- In the file **.env**, define your environment-specific configuration variables as key-value pairs.

index.js

```
require('dotenv').config();

let port = process.env.APP_PORT;
```

- In your JavaScript code, import the package.
- dotenv loads the variables into the **process.env** object, making them accessible in your code.

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cors

- CORS stands for Cross-Origin Resource Sharing, and it is a security feature implemented by web browsers.
- By default, web browsers enforce the Same-Origin Policy, which means that a web page can only make requests to the same domain (origin) from which it was served.
- **cors** package provides a middleware that can be used to enable cross-origin requests with various options.

index.js

```
const cors = require('cors');

// Enable CORS
app.use(cors({
    origin: process.env.CLIENT_URL
}));
```

npm i cors

- The "**app.use**" function is used to mount middleware in Express, and in this case, it's adding the **cors** middleware to all routes.

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Summary

1. Explain the role of Node.js in Full Stack Development.
2. Define what a RESTful API is.
3. Discuss how to use Express.js for building web APIs.
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Credits

This learning unit was developed by:

Li Haiyun, School of Information Technology, Nanyang Polytechnic

References:

- Coding stickers created by kerismaker - Flaticon
  https://www.flaticon.com/stickers-pack/web-development-195

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